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The Completion

****A relativistic field theory carrying $a_0 = \kappa c \sqrt{(G\rho_\Lambda)}$ ****

Zimmerman, Carl P. — Briar Creek Tech Assembled 2026-08-09; **v2, v3, v4, v5 same day; v6 2026-08-10; v7 2026-08-10 (late). v7 changes — the largest POSITIVE structural change of the programme: $a_0(z)$ is DERIVED from the action.** Until v7 the redshift scaling of a_0 was imposed on the theory as a phenomenological law dressed in DESI's CPL parameters — a dressing that was **never self-consistent** with this paper's own $w = -1$ exact (§1.2), and is here withdrawn and replaced. The derivation (new §1.3, rows 18–19): promote the constant a_0^2 to $a_0^2(\mathbf{Q}) = \kappa^2 c^2 \mathbf{G}(-\mathbf{K}(\mathbf{Q}))/c^2$ — **the MOND scale is the dark sector's PRESSURE** ($p = K$ is an identity in this sector). Zero new parameters; ONE new relation $\mu^2 \Lambda D^2 = M^4$ **** (“the Lagrangian vanishes at the DBI wall”), which removes a free parameter (§3: 5 → 4). Today $-K = \rho\Lambda$: the committed coefficient to the digit. Into the past the trace excitation's positive pressure climbs the DBI wall and cancels the vacuum's negative pressure: MOND switches off at recombination as an output****, with $w = -1$ still exact — the vacuum never rolls; the *total pressure* evolves. An adversarial external pass (Gemini) proved the density promotion ($a_0^2 \propto \rho Q$) *yields $a_0 = \text{const}$ or the wrong sign; that theorem is what forces the pressure route, and is credited in row 18. The full perturbation matrix with the new YQ mixing is DONE* (row 19): the cosmological matrix is untouched *exactly*, the no-ghost theorem **transfers** kernel-independently, $c_T = 1$ survives. Also in v7: the forest-sector correction — stage 15's claim that the Hooper et al. generic-shape likelihood is public was **WRONG** (announced, never released; all probes 404); the sector's forest suppression is instead **self-computed** by a lognormal mock (0.20% at the binding bin, nuisance-dominated, **stage16**). Still owed after v7: the CLASS re-run with the derived law, the MUSE/MSA-3D re-exam against the bumpless shape, and a *derivation* of $\beta = 1$ (it is **selected**, not derived — non-claim 2e). **v6 changes — an ERRATUM against v5, filed against my own argument.** v5's non-claim 2d rested one of its three steps on a **category error**: it bounded the *export* of the dust's energy from a galaxy by the khronon's own polytropic sound speed (≈ 690 Gyr). $\nabla_\mu T^{\hat{\mu}\nu} = 0$ *bounds the total energy, not the flux velocity; the bound on how fast energy can leave a region is causality, and AeST has two massless tensor modes at exactly c — by this paper's own row 17. An $O(c)$ export channel therefore exists in the theory as written (1 Mpc in 3.3 Myr). That step is withdrawn. Also withdrawn: the claim that the leak rates “lock” at $\rho\Lambda/\rho_{\text{dust}} = 2.60$ — that is merely today's density ratio. The conclusion survives on stronger, published, transport-independent grounds: exporting the energy removes it from the galaxy, not the universe, so the measured global budget decides — a halo-gated conversion needs a converted fraction $\zeta \approx 0.6$ against the **epoch-marginalised** bound $\zeta < 0.0204$ (Planck) to 0.0374 (+lensing+BAO+SN+DES) [McCarthy & Hill, PRD **108**, 063501], over by 16–29×, and would leave $\Omega_{dr,0} \approx 0.196 \approx 2100\times$*

today's radiation density; the dust→dark-energy channel needs far more than the published $|\delta_0| \leq 6.7 \times 10^{-4}$. And the halo gate is **adverse**, not protective: dilution goes as a_{-c} , so later injection leaves more residue. **The lensing theorem ($\rho+3p$ vs $\rho+p$) is untouched and is what carries the result.** Banked as a genuine asset: the Y-gate's FRW exactness makes recombination-era N_{eff} immunity exact. Also recorded, against expectation: the **background** does not require dark matter at all (Kunz's dark degeneracy, confirmed numerically — a single free $w_X(z)$ reproduces Λ CDM's $H(z)$ with baryons alone); the binding constraint is **CMB lensing**, which forces the clustering to survive to $z < 0.30$. Every numbered property below is backed by a committed, runnable script that exits non-zero on failure. Scripts named at each line. **v5 changes** — **the largest revision, and it runs against the programme's interest.** A six-stage sequence (**nbody_2026/**, every stage committed and green) settles the galaxy-interior question that v3 reopened as non-claim 2d, and **settles it adversely: the Q-sector cannot also be absent from galaxies, and the repair space of local, energy-preserving modifications is provably empty.** Three results carry it: (i) the dust mass IS the conserved shift charge ($\rho = Q_0 n$ with $\rho/n = Q_0$ regardless of any added Y-dependent Q-mass), so it cannot be suppressed locally — which retro-explains three withdrawn mechanisms; (ii) unlocking the charge does not unlock the energy ($\nabla_\mu T^{\mu\nu} = 0$ holds regardless — though v6 withdraws the transport step that accompanied this, see the v6 banner); (iii) ** dynamics responds to $\rho + 3p$ but lensing responds to $\rho + p$, and no equation of state kills both** — so the same no-slip identity that underwrites row 1 forbids hiding the dust. Also corrected here: v3/v4's claim that the DBI cap bounds the dust *density* (it bounds the pressure; ρ_{exc} is linear in u and diverges at saturation), and v4's spliced amplitude band. **Rows 1–12 and 15–17 are untouched.** **v2 changes:** row 13's shift-charge cluster route is **withdrawn** (killed by the 1-Mpc confrontation: smooth accretion restores $\$ \rightarrow 1$ for any cold IC) and replaced by the $* a\{0\}$ —bump environment response $*$, which passed a five—environment matrix AND its perturbation health check; the health check carved two hard constraints $\leq 8.4 \times 10^{-7}$; amplitude $\leq 2.7 \times$ fiducial). The coefficient row now reads as **measured:** $\kappa = 0.551 \pm 0.043, \pm 0.063$ if the H_0 tension is carried as a systematic ($\kappa \propto a_0/H_0$). **v3 changes:** (i) the **full AeST perturbation matrix** for the bump term is now DONE and added as row 17 — $c_T = 1$ exact, no-ghost as a theorem, quasi-static closure derived, amplitude cap softened to a band; non-claim 2c is updated accordingly. (ii) **Non-claim 2 is corrected:** v2's “by item 13 it is absent where rotation curves are measured” was a dangling pointer — row 13 was withdrawn *in the same version* — so the galaxy-interior status of the Q-sector dust is **REOPENED** and now stated as its own open problem (non-claim 2d). v2 carried this internal inconsistency for several hours; the correction runs against the framework's interest and is flagged rather than hidden. (iii) §3's amplitude cap now states its units ($4.5 \text{ Mpc}^{-2} = 2.7 \times$ the fiducial 1.65 Mpc^{-2}).

1. The action

$$S = \int d^4x \sqrt{-g} \left[\underbrace{\frac{R - 2\Lambda_{\text{bare}}}{16\pi G}}_{\text{Einstein-Hilbert}} + \underbrace{\mathcal{L}_{\text{aether}}[A^\mu, g]}_{\text{unit-timelike } A^\mu A_\mu = -1} + \underbrace{\mathcal{F}(Y, Q)}_{\text{the dark sector}} \right] + S_{\text{matter}}[g, \psi]$$

with the two scalar invariants built from the khronon φ and the aether A^μ ,

$$Q \equiv A^\mu \nabla_\mu \varphi, \quad Y \equiv (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi.$$

_aether is AeST's FULL aether sector, written out because v7's SVT campaign (row 20) proved that leaving it schematic invites silent truncation:

$$\mathcal{L}_{\text{aether}} = \frac{1}{16\pi G} \left[-\frac{K_B}{2} F^{\mu\nu} F_{\mu\nu} + 2(2-K_B) J^\mu \nabla_\mu \varphi - (2-K_B) Y + \lambda (A^\mu A_\mu + 1) \right], \quad J^\mu = A^\nu \nabla_\nu A^\mu$$

The two **(2-K_B) scalar-coupled terms are part of AeST's kernel** (SZ Eq. 5) and are load-bearing — they generate the committed quasi-static $G = (1-K_B/2)\hat{G}$, and omitting them changes the scalar and vector spectra (row 20 caught exactly this in an intermediate derivation and killed the artifacts).

This is **Aether-Scalar-Tensor** (Skordis & Złośnik 2021, PRL **127** 161302) — chosen because it is the only relativistic MOND-class theory that reproduces the CMB. The framework's content is entirely in the choice of the free function , which **factorises**:

$$\boxed{\mathcal{F}(Y, Q) = \underbrace{\frac{a_0^2}{8\pi G} \mathcal{F}_Y\left(\frac{Y}{a_0^2}\right)}_{\text{MOND}} + \underbrace{K(Q)}_{\text{dark energy + dark matter}} + \underbrace{A B\left(\frac{Y}{a_0^2}\right) (Q - Q_0)^2}_{a_0\text{-bump response (v2)}}$$

$$B(y) = \frac{y}{(1+y)^2}$$

The third term (v2) is a **position-dependent Helmholtz mass** $\mu_{\text{eff}}^2 = A B(g^2/a_0^2)$ **peaked at the framework's own** a_0 — the location costs nothing (Y/a_0^2 is already the Y-sector's normalisation); one new calibrated amplitude $A \approx 1.7 \text{ Mpc}^{-2}$. *The dark sector's response is resonant at the MOND transition, and clusters are the cosmic objects that live at a_0 (R_{500} at 0.33–0.58 a_0).*

1.1 The Y-sector — MOND

Y is fixed by the framework's Route A kernel $\nu(y) = 1/(1 - e^{-\sqrt{y}})$, $y = g{\text{bar}}/a_0$. In AQUAL variables it is the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \quad x(u) = \frac{u^2}{\mu(u)}, \quad u = \sqrt{y}, \quad x = g_{\text{obs}}/a_0.$$

with the normalisation

$$a_0 = \kappa c \sqrt{G \rho_\Lambda} = \frac{c H_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.3619 \times 10^{-11} \text{ m s}^{-2}, \quad \kappa = \frac{1}{2}, \quad Z = 2\sqrt{8\pi/3}.$$

$\kappa = \frac{1}{2}$ is **FITTED, not derived**. See §4.

1.2 The Q-sector — an offset DBI khronon

$$K(Q) = -M^4 + \mu^2 \Lambda_D^2 \left[1 - \sqrt{1 - \frac{u^2}{\Lambda_D^2}} \right], \quad u \equiv Q - Q_0, \quad |u| < \Lambda_D$$

Three jobs from one function:

piece	limit	what it is
the offset $-M^4$	$u = 0, K = 0$	$\rho = -K = M^4, p = K = -M^4$ $\Rightarrow \mathbf{w} = \mathbf{-1}$ exactly : dark energy
the deviations	$u \ll \Lambda_D$	$K \rightarrow \mu^2 u^2 / 2$ is the pressure , while $\rho_{\text{exc}} = Q_0 \mu^2 u$ is linear in $u \Rightarrow w = u / 2Q_0 \rightarrow 0$: dust , i.e. the dark matter. (<i>v5: the linearity is why the dust mass equals the conserved charge — see non-claim 2d.</i>)
the saturation	$u \rightarrow \Lambda_D$	K bounded, $K \rightarrow \infty \Rightarrow w \rightarrow 0$ again: kills the early stiff phase

Parameters: M (vacuum scale, $M^4 = \rho_{\text{vac}}$), μ (Helmholtz mass), Λ_D (DBI field-space scale), I_0 (the conserved shift charge = the dust amount). *(v7: Λ_D is no longer free — $\mu^2 \Lambda_D^2 = M^4$, §1.3.)*

1.3 $a_0(z)$ — derived, not imposed (v7)

The constant a_0^2 in §1.1 is promoted to a scalar of the Q-sector:

$$a_0^2(Q) = \frac{\kappa^2 c^2 G(-K(Q))}{c^2}, \quad \text{i.e.} \quad a_0 = \kappa c \sqrt{\frac{G(-p_Q)}{c^2}}$$

The MOND scale is the dark sector's pressure — and since $p = K$ identically (§1.2), this is the unique algebraic scalar of Q that (i) equals ρ_{vac} today ($u \rightarrow 0$: $-K = M^4$, reproducing §1.1's normalisation to the digit) and (ii) *declines* into the past. The competing candidate, $a_0^2 \propto \rho_{\text{exc}}$, is **excluded by one bit**: the density grows with excitation, so a_0 would rise toward recombination — MOND ON at the CMB (the adversarial theorem credited in row 18 proved that route also gives $a_0 = \text{const}$ on the pure-vacuum branch; either way it is dead). The pressure route additionally has charge-suppressed backreaction: $da_0^2/dQ \propto -K = -n$, versus the density route's QK , unsuppressed.

With $\beta \mu^2 \Lambda_D^2 / M^4 = 1$ — the Lagrangian vanishes exactly at the DBI wall — the background law is closed-form ($n \propto a^{-3}$ exactly; $\nu n / (\mu^2 \Lambda_D)$):

$$\frac{a_0^2(z)}{a_0^2(0)} = \frac{\sqrt{1 + \nu_0^2}}{\sqrt{1 + \nu_0^2(1 + z)^6}}$$

constant to $<1\%$ through every MOND test ($z \leq 5$), transition at $z_t = \nu_0^{-1/3} - 1 \in [17, 35]$, off at recombination (0.002–0.006). **w = -1 stays exact throughout** — the vacuum never rolls; the *trace* excitation ($\Omega \leq 4.4 \times 10^{-7}$, the stage-3 black-hole ceiling) climbs the wall and its pressure reaches M^4 regardless of its amount, because the wall height is a property of the potential, not the charge. The window $\nu_0 \in [2.1 \times 10^{-5}, 1.8 \times 10^{-4}]$ is cut from *below* by the CMB off-switch and from *above* by the RAR — and the RAR ceiling exists only because the captured charge **drains** (stages 2–3): had it formed a halo, the local a_0 would fall to 0.55 of nominal at the $g = a_0$ radius, RAR-fatal. **The adverse drain result of non-claim 2d is load-bearing for this derivation.** The old CPL-dressed law (bump at $z \approx 0.5$) is withdrawn; the derived law has no bump. Scripts: `nbody_2026/stage17_a0z_from_the_action_2026.py` (17/17), `nbody_2026/stage18_perturbation_matrix_with_mixing_2026.py` (10/10).

2. What is verified

#	property	value / statement	script
1	Lensing — the test that killed modified inertia	$\Phi = \Psi \Rightarrow \gamma_{\text{PPN}} = 1$, $M_{\text{dyn}}/M_{\text{lens}} = 1$ exactly. $21.2\sigma \rightarrow 0.601\sigma$	<code>mi_relativistic_completion_aest.</code>
2	Kernel embeds in $_{\text{Y}}$	deep-MOND $\mu \rightarrow x$ exact; Newtonian $\$ \rightarrow 1; x(u) \text{abijection}(h = e^{(-u)}(1+u) > 0); \text{free function} * \text{convex} * * \text{same} 3 * \text{Newtonian residual} * e^{(-\sqrt{\$}y)} = 3.6 \times 10^{-3457}$ at Earth's orbit	same
4	Dark energy is not an input	$w = -1$ exactly at the condensate minimum — proved, not assumed	<code>mi_condensate_vacuum_energy_a0_2</code>
5	Ghost-free over the whole field range	$K = \mu^2(1-s^2)^{-3/2} > 0$ for all $ s < 1$	<code>mi_dbi_khronon_2026.py</code>
6	Subluminal everywhere	$c_s^2 = \Lambda_{\text{D}}$ $s(1-s^2)/(1+\Lambda_{\text{D}} s) \leq 0.385 \Lambda_{\text{D}}$	same

#	property	value / statement	script
7	The published no-go is dissolved	Blanchet & Skordis 2024 §4.3.1's $455 \times$ conflict reverses sign : bounded pressure $\Rightarrow w \rightarrow 0$ early $\Rightarrow$ cosmology becomes a <i>lower</i> bound on μ^{-1} , same direction as MOND	same
8	Theorem : why no polynomial works	for $K \sim u^n$, early-time $w \rightarrow 1/(n-1)$. Quadratic 1, quartic $\frac{1}{3}$, sextic — all fail. Only boundedness gives $w \rightarrow 0$	same
9	CMB acoustic peaks — real CLASS run	at $\Lambda_D \leq 10^{-2}$, $c_s^2(\text{recomb}) =$ 2.9×10^{-8} ; TT unchanged to 0.069% , $P(k=0.2)$ to 1.71%. Indistinguishable from CDM	mi_dbi_cmb_class_run_2026.py
10	BTFR	exact, from convexity	mi_route_a_field_theory_2026.py
11	g^{-2} Lorentz violation	restored by the aether (pure Bekenstein–Milgrom had lost it)	mi_relativistic_completion_aest.
12	a_0 tied to Λ	$a_0 = m_{\text{cond}}/(4\sqrt{\pi})$ with $m_{\text{cond}} =$ $M^2/(\sqrt{2}M_{Pl}), M^{\wedge\{4\}}$ $= \rho_\Lambda$ — agrees to 0.076%	mi_condensate_vacuum_energy_a0_2

#	property	value / statement	script
13	Clusters (v2)	shift-charge IC route withdrawn: killed by the 1-Mpc confrontation — cluster R500 and galaxy-outskirt lensing share a scale with disjoint requirements, correlated Gaussian ICs cannot select environments, and smooth accretion restores \$ → 1 for any cold IC * *. Replaced by the * * a {0} — bump response * *(row15) mi_c,oute_1 mpa auto-fiduciation_2026.py` (9/9) 14 * * Directional EF E signal * * pure MI predicted * exactly zero * ; AQUAL — class predicts 1–4 15 * * a {0} \$-bump environment response	passes all five environments: cluster (calibration, $\mu^2_{\text{eff}} = 0.23 \text{ Mpc}^{-2}$), galaxy interior 0.034% of M_b (RAR cost 4×10^{-4} dex), galaxy 1 Mpc 1.2% vs the 5.9% strict bound, linear cosmos 0.6% of mean matter, solar 10^{-19} . Vanishes on FRW (Y=0) and is second-order in perturbations $\Rightarrow$ linear CMB/P(k) construction. Evades both prior kills (a response has no charge to advect; couples to environment directly)
16	...and its health check	no ghosts ever (kinetic contribution $2AB \geq 0$, $= 0$ on FRW), no Ostrogradsky sector; FRW gradient health **excludes the old $\Lambda_D = 10^{-2}$ and forces $\Lambda_D \leq 8.4 \times 10^{-7}$ (window stays 3.7 orders); halo gradient health caps $A \leq 2.7 \times$ fiducial, excluding the demanding end of Mistele's cluster band — a two-sided falsifiable pinch; free signature: cluster anisotropic stress at $O(0.6)$	mi_a0_bump_health_2026.py (11/11)

#	property	value / statement	script
17	...and the FULL perturbation matrix (v3)	with all aether mixings: c_T = 1 exact (the new terms are algebraic in the metric — GW170817 safe by construction); no-ghost is a THEOREM (the bump’s kinetic addition is rank-1 PSD in the $\chi\chi$ entry $\Rightarrow$ Weyl monotonicity: it cannot lower any eigenvalue of AeST’s healthy kinetic matrix; closed-form $2\times 2 + 200$ random 4×4); unit-norm gives $\delta A^0 = -\Phi A^0 \Rightarrow \delta Q = \chi - Q_0\Phi$ — the quasi-static phenomenology of row 15 is derived, not assumed; gradient block exact: $\Delta G = 2\lambda[[q, 2y^{\{3/2\}}B], [2y^{\{3/2\}}B, yq]]$, $q = (3y^2-8y+1)/(1+y)^4$; integrating out the aether softens the amplitude cap: at $\text{base_a} = 1$ the band is A_max $\in [2.72, 4.46] \times \text{fiducial} = [4.49, 7.36] \text{ Mpc}^{-2}$ (v4 correction: the earlier “(2.7–7.4) $\times$ fiducial” spliced fiducial units with Mpc^{-2}, making the band look $1.66\times$ wider at the top than it is — base_a_lookup_and_robustness_2026.py, 12/12). The base_a lookup is now DONE and it	mi_aest_full_matrix_bump_2026.py (11/11)

#	property	value / statement	script
18	a₀(z) DERIVED from the action (v7)	$a_0^2(Q) = \kappa^2 c^2 G(-K(Q))/c^2$ — the MOND scale is the dark sector's pressure. Zero new parameters; one relation $\beta =$ $\mu^2 \Lambda_D^2 / M^4 = 1$ (selected) by the CMB off-switch, not derived — non-claim 2e). Today's coefficient reproduced to the digit; w = -1 stays exact (the vacuum never rolls; the total pressure evolves). Density promotion excluded by one bit (a_0 would rise into the past); the exclusion theorem for the naive route is due to an adversarial external pass (Gemini, 2026-08-10) and is credited as such. Closed-form law: $a_0^2(z)/a_0^2(0) =$ $\sqrt{(1+\nu_0^2)}/\sqrt{(1+\nu_0^2(1+z)^6)}$; constant to <1% for z ≤ 5; transition $z_{\text{t}} \in$ $[17, 35]$; off at recombination (0.002–0.006). Window $\nu_0 \in$ $[2.1 \times 10^{-5},$ $1.8 \times 10^{-4}]$, cut by the CMB from below and the RAR-via-drain from above — non-claim 2d's adverse result is load-bearing here. The CPL-dressed law (never self-consistent with $w = -1$ exact) is 9 withdrawn	stage17_a0z_from_the_action_2026-08-10 (17/17)

#	property	value / statement	script
19	...and ITS full perturbation matrix (v7)	the promotion's <code>_YQ</code> $\neq 0$ mixing analysed: the cosmological matrix is untouched EXACTLY ($h^{00} =$ $g^{00} + A^0 A^0 = 0$ exactly for $A^i = 0$ — with the aether scalar on, $h^{00} =$ $(\partial\alpha)^2/a^2$ at <i>second</i> order (row 20 refinement); $\delta Y^1 = 0$ identically either way; every new entry carries a positive power of y, and $y = 0$ on FRW — the CLASS pass, k^4-Jeans structure and GDM degeneracy theorem inherit without recomputation); the no-ghost theorem TRANSFERS kernel- independently (both kinetic additions land in the same $\chi\chi$ slot as row 17's: $(F\cdot yF) \geq 0$ because ghost-freedom of the base meets concavity of the kernel, and $(^2/y)^2F$ is a perfect square with $\propto -K = M^4/\sqrt{(1+\nu^2)}$ > 0 always — a_0^2 can never cross zero; Weyl monotonicity does the rest); the mixing never reaches the kinetic matrix (δY has no time derivative), its drift dispersion is real at every k, and its gradient/force corrections are 10trace-bounded at $1.3 \times 10^{-4}/1.7 \times 10^{-4}$ of committed entries; $c_T = 1$ survives	<code>stage18_perturbation_matrix_with</code> (10/10)

#	property	value / statement	script
20	...and the COVARIANT SVT DECOMPOSITION (v7, post-publication)	nine-agent derive-and-adversarially-verify campaign; 17 executed sympy scripts committed in nbody_2026/svt_2026/, all exit 0. CONFIRMED under the full action: $c_T = 1$ exact from explicit $O(h^2)$ components $((a^3/32\pi G)^2 - (a/32\pi G)(\partial h)^2$ per polarisation; gravitons massless on the background shell; independent Euler route agrees); the promotion drops out of FRW perturbations at second order — the whole promoted MOND term starts at THIRD order (verified with generic (Q) and live chain rule); $\delta A^0 = -\Phi + (3/2)\Phi^2 + (\partial\alpha)^2/2a^2$ exact; $\delta Y^2 = \partial(\chi+Q\alpha) ^2/a^2$, a perfect square in the shift-invariant combination; the quasi-static $\chi\chi$ block exact (unchanged by $\sqrt{(-g)}$ and second-order $\delta Q/\delta Y$ pieces); drift asymmetry closed form $D^2/(G_KG_G)$ = $(\kappa^2/4\pi)s^2y^3F^2/(F+2yF)$, ceiling 5.1×10^{-3}; corrected coefficients: second kinetic addition $\leq 0.0267\kappa^2s^2K$, q-window (0.1315, 2.5352). THE	stage22_covariant_svt_decomposition (20/20, drives the full suite)

3. Parameter count, honestly

	Λ CDM	this completion
dark sector	$\Omega c, \Omega \Lambda$	$M (= \rho_- \Lambda^{\{1/4\}}), I_0, \mu,$ $\Lambda_D^{**}(\text{v7: } \Lambda_D = M^2/\mu, \text{ no}$ longer free), $A (\text{v2})^{**}$
MOND scale	—	a_0 — not independent , = $m_cond/(4\sqrt{\pi})$ via item 12; (v7) and its z-dependence is now derived too (§1.3)
bounds	—	$\Lambda_D: 1.9 \times 10^{-10} \ll \Lambda_D \leq$ 8.4×10^{-7} (health-bounded, v2; 3.7 orders) — (v7) now fixed at M^2/μ by $\beta = 1$, which must land inside this window: a consistency check inherited rather than retired; $A \approx 1.7 \text{ Mpc}^{-2}$ calibrated on clusters, health cap $A \leq 4.5 \text{ Mpc}^{-2} = 2.7 \times$ fiducial (row 16; the full matrix widens it to $[2.72,$ $4.46] \times \text{fiducial} = [4.49, 7.36]$ Mpc^{-2} at $\text{base_a} = 1$ — row 17, where the base_a fork and its robustness are stated); μ^{-1} $= 4392 \text{ Mpc}$ (item 12); $I_0 \approx$ Ω_dm (an IC); (v7) the khronon's own trace charge $\nu_0 \in [2.1 \times 10^{-5},$ $1.8 \times 10^{-4}]$ (§1.3)

So: **(v7) four dark-sector numbers against Λ CDM's two** (v2 added A ; v7 removes Λ_D via $\beta = 1$), with a_0 — value *and* redshift dependence — derived from them rather than added. Still not fewer parameters than Λ CDM. What it buys is that Λ , dark matter, MOND, *and now the evolution of the MOND scale* come from **one function** instead of unrelated sectors.

4. What is NOT claimed — read this before quoting anything above

1. $\kappa = \frac{1}{2}$ **IS NOT DERIVED**. Item 12 looks like a derivation and is not. $a = m_cond/(4\sqrt{\pi})$ is *algebraically identical* to $a = \sqrt[4]{G_A}$ and to $Z = 2\sqrt{8/3}$ — three costumes, one statement — and the residue is **convention-dependent** ($4\sqrt{\pi}$ with the reduced Planck mass,

with the non-reduced). A derived number cannot depend on a bookkeeping choice. It is a **relabelling**, as this corpus's own theorem already established. What genuinely moved: "why $\kappa = \frac{1}{2}$?" becomes "why is $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$?", which is sharper because m_{cond} is a *derived* quantity of a real field theory rather than a definition.

2. **Dark matter EXISTS**, at the full Ω_{dm} . Removing the pressureless component moves H_3/H_1 by 54% and no refit absorbs it ($\Delta\chi^2 > 400$). What does **not** exist is a dark-matter *particle*: it is the Q-sector of the same scalar that supplies Λ and MOND. (**v3 correction**: v2 continued "and by item 13 it is absent where rotation curves are measured" — a dangling pointer, since v2 itself withdrew row 13. See 2d. **The defensible slogan shrinks to "no dark-matter particle" — never "no dark matter"; and (v5) "none in galaxies" is WITHDRAWN as a claim of this theory, not merely open — see 2d.)** 2d. (**v5) THE Q-SECTOR CANNOT ALSO BE ABSENT FROM GALAXIES — FALSIFIED WITHIN THE THEORY AS WRITTEN, AND THE REPAIR SPACE IS NOW PROVABLY EMPTY.** This is the largest change since v1 and it runs against the programme's interest. v2 asserted "none in galaxies" via a pointer to withdrawn row 13; a six-stage sequence (nbody_2026/, all scripts committed, 20/20 + 15/15 + 12/12 + 15/15

- 15/15 + 11/11) then settled the question the pointer had concealed.

The physics. The 1-Mpc smooth-accretion theorem implies the Q-sector dust falls into *every* collapsing basin, galaxies included. What happens next was the open problem, and it closed in four steps:

- **Nothing stops the collapse.** Wave (k^4 ghost-condensate) pressure gives a soliton scale of **0.18 AU** at halo density — eight orders below the RAR region — and it *shrinks* as ρ grows ($\lambda \propto \rho^{-\frac{1}{4}}$); a 1 kpc wave core would need M below the Lyman- α fuzzy-DM floor. The dust *is* an $n = 1$ polytrope ($p = K\rho^2$, $c_s^2 = 2K\rho$ rising) with a genuine **mass-independent 105 pc** equilibrium — but that lies **14.4× past DBI saturation**, where $c_s \rightarrow 0$ and the stiffening switches off. Endpoint: a black hole of the captured share, **falsified** $5.8 \times 10^5 \times$ **against Sgr A*** ($4.3 \times 10^6 M$ from individual stellar orbits — no mass model, kernel or cosmology in the comparison). *(v5 corrects v3/v4 here: an earlier draft claimed the DBI cap bounds the density and yields a ~ 250 pc core. $\mu^2 u^2/2$ is the **pressure**; exactly, $\rho_{\text{exc}} = Q_0 \mu^2 u$ is **linear** in u and diverges at saturation. Withdrawn.)*
- **THEOREM: the dust mass IS the conserved shift charge.** $\rho = Q_0 \cdot n$ with n the shift-charge density, and $\rho/n = Q_0$ **independently of any Y-dependent Q-mass, of any shape or amplitude.** So no such term can change how much dust a galaxy carries — *a conserved charge cannot be suppressed locally, only moved (gravity moves it inward) or not conserved.* This single statement retro-explains three withdrawn mechanisms: the shift-charge IC route, the Φ^n response, and an acceleration-gated suppressor built and killed in stage 4–5 (its eight gates all tested the field amplitude u ; the observable is n , and at fixed charge the suppressor makes a region a *cheaper* place to store charge — it attracts what it was built to expel).
- **Breaking the shift symmetry gets furthest, and fails.** A *global* potential is incompatible as an identity ($dV/dt = V\varphi$ must vanish for $w = -1$, while the condensate needs $\varphi = Q_0 \neq 0$); quantitatively the leak rates lock at $\Gamma n/\Gamma V = \rho\Lambda/\rho_{\text{dust}} = 2.60$, so clearing a galaxy in 1 Gyr drifts $\rho\Lambda$ *by 47× and protecting* $\rho\Lambda$ forces a 180 Gyr charge lifetime. A **Y-gated** breaking, $-W(Y/a_0^2)V(\varphi)$ with $W(0) = 0$, *defeats* that — on FRW the symmetry is **exact**, so cosmology is untouched identically. But **unlocking the charge does not unlock the energy**: $\nabla_\mu T^\mu{}_\nu = 0$ holds regardless. (**v6 erratum**:

v5 added that the energy therefore *cannot leave*, bounding its export by the khronon's polytropic sound speed at ≈ 690 Gyr. That was a category error — a conservation law bounds total energy, not flux velocity; the causal limit is c , and row 17's tensor modes travel at exactly c . **Withdrawn.** What kills the export route instead is the *measured global budget*: a halo-gated conversion needs $\zeta \approx 0.6$ against the epoch-marginalised $\zeta < 0.0204\text{--}0.0374$ [McCarthy & Hill, PRD **108**, 063501] — over by $16\text{--}29\times$ — leaving $\Omega_{\text{dr},0} \approx 0.196$, some $2100\times$ today's radiation density; and dust $\rightarrow \Lambda$ needs far more than $|\delta_0| \leq 6.7 \times 10^{-4}$. The gate is adverse rather than protective, since dilution goes as a_{c}^{**})

- **** AND LENSING CLOSES IT BY A THEOREM.**** Transforming the equation of state almost works: the dynamical source $\rho + 3p$ vanishes at exactly $f = 1/3$, and the field-strength gate makes that a *stable fixed point* rather than a tuning. But **dynamics responds to $\rho + 3p$ while lensing responds to $\rho + p$** — at $f = 1/3$ the lensing source is still $(2/3)\rho$, giving $M_{\text{lens}}/M_{\text{dyn}} \approx 29$ against an observed $1.0\text{--}1.3$. Generally, $\rho + 3p = 0$ requires $w = -1/3$ and $\rho + p = 0$ requires $w = -1$: **incompatible. No equation of state renders a given energy density invisible to both orbits and light; the only such configuration is $\rho = 0$** , which the transport bound forbids reaching. *The no-slip identity $\Phi = \Psi$ that makes this theory's modified-gravity arm work — the property that killed its modified-inertia arm at 21σ and underwrites item 1 — is the same property that forbids hiding the dust.*

Status, stated without spin. The galaxy-interior problem is not a missing mechanism: it is a conservation law (energy, via general covariance), plus a transport bound, plus lensing. **The repair space of local, energy-preserving modifications is provably empty.** The defensible slogan is therefore “**no dark-matter particle**” and nothing more; “**none in galaxies**” is **withdrawn as a claim of this theory**. What remains open is a structural change — the Q-sector not acquiring the galactic energy in the first place — against which the smooth-accretion theorem stands for any cold initial condition. Items 1–12 and 15–17 are untouched by all of this.

2e. (v7) **THE $a_0(z)$ DERIVATION'S OWN CAVEATS, stated before anyone quotes row 18 as finished.** (i) $\beta = 1$ is **BOUNDARY-PINNED** (*upgraded 2026-08-10 from “selected” by `nbody_2026/stage20_beta_equals_one_derivation_2026.py`, 8/8*). Three results: $\beta \leq 1$ is a **THEOREM** — an additive constant in K is Λ_{bare} in disguise (a redundancy the promotion breaks), and any $\beta > 1$ makes a_0^2 cross zero at finite redshift, where the kernel goes complex for any spatial khronon gradient ($\beta = 1.01$ dies before recombination; $\beta - 1 = 10^{-8}$ by $z \sim 10^4$); $\beta = 1$ **K is a PURE brane action** (the offset-DBI decomposes exactly into tension $\times$ volume element + constant; the constant-free form's zero at the wall is canonical — a null brane has zero action); and the CMB pins $1 - \beta \leq 3.6 \times 10^{-5}$ from below. The defensible sentence: “ $\beta \leq 1$ is derived; the boundary is canonical under the brane reading; the CMB pins the theory to within 3.6×10^{-5} of it.” Anyone quoting “ $\beta = 1$ is derived” is overquoting. Named escapes, kept open: $\beta = 1 - \varepsilon$ is consistent with a falsifiable high- z MOND floor $\sqrt{\varepsilon} \cdot a_0$ (a detection would *measure* $1 - \beta$); and the $\beta \leq 1$ theorem needs the charge history to extend to arbitrarily early times. (ii) **The DESI tension is not resolved and cannot be resolved by this sector**: the non-dust remainder's $w(z) \geq -1$ always, so DESI's preferred phantom past ($w_0 + w_a = -1.61$) is unreachable. If DESI's evolving- w detection stands, it remains evidence against $w = -1$ exact. What v7 fixes is only that the $a_0(z)$ law no longer *depends* on borrowing DESI's numbers. (iii) ~~The CLASS re-run with the derived law~~ — **CLOSED 2026-08-10 by `nbody_2026/stage19_class_rerun_derived_law_2026.py`**

(8/8): the exact $\beta = 1$ background is $\rho nd = M^4 \sqrt{1+\nu^2}$, $p_{nd} = -M^4 / \sqrt{1+\nu^2}$ ($\rho p = -M^8$ an exact invariant) — vs Λ CDM one cold a^{-3} trace (today-density $\rho \Lambda \nu_0 \leq 1.2 \times 10^{-4} \rho_{\text{crit}}$) and an a^{-6} transient $< 3.2 \times 10^{-8}$; θ^* moves $0.18\times$ Planck precision; real CLASS runs give $\Delta\chi^2 = 1.3$ against cosmic variance over 4998 TT+EE multipoles (0.01σ — an ideal experiment could not distinguish them); the MOND-off input of the committed pass is now an output, $a_0(z^*) = 0.002\text{--}0.006$ of today. Same rigor level as the committed pass (stock CLASS, trace bracketed as cold matter — exact at high z , conservative at low z). ~~The MUSE/MSA-3D re-examination~~ — **CLOSED 2026-08-10 by nbody_2026/stage21_muse_msa_reexam_derived_law_2026.py** (7/7): the MUSE tension *persists* (15.1σ raw, $\sim 2\text{--}3\sigma$ after the shared Λ CDM assembly drift) but re-grades from framework-distinctive-WORST (the withdrawn declining branch: wrong sign, 16.3σ) to **shared-with-all-constant- a_0 -MOND**; in the committed joint fork likelihood the framework's branch relabels M-DEC $\rightarrow$ M-FLAT (the branch needing exactly the drift MSA-3D measures, $p = 1.22$, and its near-unfalsifiability critique inverts into the sharp null below); MSA-3D re-grades WEAK-TENSION/WATCH $\rightarrow$ CONSISTENT (1.15σ from flat). **Forfeited, priced:** the $z \approx 0.4$ bump defense; the $3.1\text{--}4.2\sigma$ DESI cosmology tailwind (and with it the $w \rightarrow -1$ hostage — dissolution is now the *prediction*); the $z \approx 2\text{--}3$ declining discriminator. **New exposure:** full-strength MOND through cosmic dawn ($a_0(10) = 0.99\text{--}1.00$ vs the old 0.36) — Nusser-type over-production unpriced, $z_{\text{t}} \in [17, 35]$ in the 21-cm window; the early-structure front needs its own confrontation. ~~The covariant SVT decomposition~~ — **CLOSED 2026-08-10 by row 20 (nbody_2026/stage22_covariant_svt_decomposition_2026.py, 20/20, driving 17 committed executed scripts):** every v7-specific claim survived adversarial verification under the full action; the truncated-action artifacts an intermediate derivation produced were caught and killed by the verifiers before entering the corpus. **Still owed** (base-AeST, not v7): the in-repo full-action FRW scalar spectrum + the k^4 soft-mode WATCH confronted with AeST's published stability discussion; and the cosmic-dawn confrontation opened by the stage-21 re-exam. (*The $\beta = 1$ derivation moved to (i): boundary-pinned, 2026-08-10.*)

3. **The post-recombination growth history is NOT verified.** Item 9 covers the acoustic peaks only. Holding c_s^2 at its peak value for all time *destroys* $P(k=0.2)$, so the pass depends essentially on the bump being transient (it peaks at $z \approx 189$). **A patched CLASS fluid carrying $c_s^2(a)$ is required, not optional.**
4. **(v4 rewording — the original referred to withdrawn row 13.)** The dust component's small-scale initial conditions are **not confronted with Lyman- α** . The withdrawn IC route needed a khronon transfer function $T \approx 0.33$ at $k \approx 4.5 \text{ Mpc}^{-1}$ (Λ CDM-excluded); the surviving picture (standard cold dust + the bump response) should be Ly- α -safe by construction, but that is **asserted, not computed — the confrontation remains owed item in the programme.**
5. **Cassini Q_2 is inherited** ($3\text{--}15\sigma$) and is not relieved by anything here.
6. **PPN preferred-frame parameters α_1, α_2** for this have not been computed against lunar-laser and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote only.** $\gamma_{\text{v}} = \sqrt{\nu(y_{\text{extN}})}$; the full nonlinear AQUAL-EFE solve is owed.
8. **This is not a theory of everything** and no part of it addresses the Standard Model. The 2026-06-23 retraction stands.

5. The falsifiable predictions

prediction	value	how to test
wide-binary boost	$\gamma_v = \sqrt{\nu(y_{\text{extN}})}$	Gaia DR4 , pre-registered and hash-stamped
directional EFE asymmetry	1–4%, signed	already firing at $p = 0.029$ with the right sign
cluster-to-cluster scatter in the residual	$\sigma(\xi)/\xi \approx 0.02\text{--}0.60$	existing cluster samples — (v4) separates the a_0-bump response (scatter tracks each cluster's distance from the resonance, rows 15–17) from any fixed-scale mechanism, which predicts uniformity (the two prior candidates are withdrawn)
cluster mass profile misfit	ρ_c flat, $M_c \sim r^3$, can go negative	fit clusters with the khronon profile instead of NFW
g^{-2} Lorentz violation	computable $s_{\mu\nu}$	SME sector bounds
accelerated structure formation	earlier massive objects	JWST high- z massive galaxies (McGaugh et al. 2024)
$a_0(z)$ shape (v7)	constant to <1% for $z \leq 5$, transition $z_t \in [17, 35]$, NO bump at $z \approx 0.5$	any measurement of a_0 evolving below $z \approx 5$ falsifies the derived law (the withdrawn CPL dressing predicted +6% at $z = 0.5$ — the two are now distinguishable); MUSE-type high- z kinematics, high- z RAR

6. Credit

$\nu = \sqrt{1+1/y}$ is **Milgrom 1999 PLA 253:273 eq 9**. MOND: Milgrom 1983 ApJ 270:365. AQUAL: Bekenstein & Milgrom 1984 ApJ 286:7. TeVeS: Bekenstein 2004 PRD 70:083509. **AeST: Skordis & Złóśnik 2021 PRL 127:161302** — the completion itself. Ghost condensate: Arkani-Hamed, Cheng, Luty & Mukohyama 2004 JHEP 0405:074; nonlinear: +Wiseman 2007 JHEP 0701:036. $K(Q) = \mu^2(Q-1)^2$ and the ghost- condensate identification: Verwayen, Skordis & Złóśnik 2024; **Blanchet & Skordis 2024 JCAP 11(2024)040** — including §4.3.1, the no-go this work dissolves. Quasi-static AeST: Durakovic & Skordis 2024 JCAP 04:040; Verwayen, Skordis & Boehm 2024 MNRAS 531:272. **Mistele, McGaugh & Hossenfelder 2023 A&A 676:A100** — the weak-lensing bounds that killed this program's first two cluster mechanisms and now two-sidedly bound the third (rows 15–17). GDM bound: Kopp, Skordis, Thomas & Ilić 2018 PRL 120:221102. Ly- α floor: Rogers & Peiris 2021 PRL 126:071302. RAR scatter: Desmond 2023. Accelerated structure formation: McGaugh, Schombert, Lelli & Franck 2024.

The distinctive content of this work is the coefficient — $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ with $\kappa = \frac{1}{2}$ — its embedding in $_Y$, the offset-DBI Q-sector, and items 7, 8, 9, 12 and 13. Everything else is cited above and is not mine.